

Sujatro Ganguli

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PROFESSIONAL SUMMARY

Third-Year Computer Science student specializing in AI Engineering, Machine Learning, and Data Architectures. Expert in orchestrating RAG pipelines, deploying local LLMs, and building vector indexing tools. Proven backend and systems engineering experience with a track record of delivering containerized applications and high-performance automated research software.

EDUCATION

Vellore Institute of Technology (VIT), Vellore — B.Tech in CSE (CGPA: 8.64) 2023 – Present

TECHNICAL SKILLS

Languages: Python (Expert), C++, C, Java, SQL

AI & Data: LangGraph, RAG Architectures, Vector Indexing, GraphDB (KuzuDB), NLP, OpenCV, MediaPipe

Deployment & Systems: Docker, CI/CD Pipelines, FastAPI, Django, Git, Linux Systems, SQLite

Domains: Local Model Orchestration, MLOps, Computer Vision, Semantic Search, Optimization Algorithms

EXPERIENCE

- **Web Development Intern — Terionix:** Architected a full-stack platform featuring a Django backend and custom automated data scrapers. Containerized the entire architecture via Docker and established a CI/CD deployment pipeline to a custom mobile-based staging server, demonstrating robust deployment and systems engineering capabilities. May 2026 – Present
 - **Research Intern — PRL:** Automated SHARAD radar visualization using Python data pipelines; optimized algorithms to reduce rendering latency for **3 GB of data** from hours to **sub-10 seconds**. 2025
 - **Technical Head — SEDS VIT:** Architected backend event portals for Hermes and Erebus; managed **270+ registrations** with **100% uptime** via Vercel and Cloudflare DNS. 2026
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AI & MACHINE LEARNING PROJECTS

- **Vessel:** Local-first, highly configurable knowledge workspace featuring an offline LangGraph-powered RAG pipeline for on-device semantic querying of study materials. *Tech: Python, LangGraph, Qt/QML, SQLite, RAG* 🏠
 - **EVA — RAG Tool:** Engineered custom .eva format for versioned vector indexing, reducing vector re-indexing time from hours to seconds for large document sets. *Tech: Python, RAG Architecture, Vector Indexing, LLMs* 🏠
 - **Arabella:** AI-powered search tool utilizing a Graph Database to extract, store, and retrieve semantic connections across unorganized text data. *Tech: Python, KuzuDB, GraphDB, Semantic Search, AI* 🏠
 - **Automated Question Gen:** RAG NLP pipeline automating academic assessment creation from source text, reducing manual formulation effort by 75%. *Tech: Python, NLP, Prompt Engineering, LLMs* 🏠
 - **Gestualis:** Low-power computer vision gesture library using complex dot-product mathematical logic to achieve 100% consistent hand-tracking. *Tech: Python, OpenCV, Computer Vision, MediaPipe* 🏠
 - **Sentimental Text:** Modular CLI tool for deep learning sentiment analysis featuring custom data preprocessing and feature extraction pipelines. *Tech: Python, Deep Learning, NLP, Data Pipelines* 🏠
 - **Exoskeleton Optimization:** Applied NSGA-II algorithms via MATLAB to optimize biomechanical parameters in real-time. *Tech: MATLAB, NSGA-II, Algorithmic Optimization* 🏠
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AWARDS & ACHIEVEMENTS

- **1st Place — Caterpillar Hackathon (2024):** Built technician app featuring real-time face-ID and voice-to-text NLP processing.
 - **Track Prize — Yantra Central Hackathon (2025):** Developed machine learning genomic analysis platform achieving **89% accuracy**.
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CERTIFICATIONS & INTERESTS

Certifications: DELF B1 (French), Udemy Python Programming (64.5 hours)

Languages: English (Professional), French (Conversational), Hindi, Bengali

Interests: Local AI Model Orchestration, MLOps, Linux System Customization